

Autonomous Evolution Standard (AEVS)

OriginID: OOF-OID-ASGA-AEVS-2026-07-08-0010
Architecture Ecosystem: Structured Reality Standards™
Architecture Family: Autonomous Systems Governance Architecture (ASGA™)
Operational Layer: Autonomous Operational Governance Layer
Category: Governance & Enforcement
Subcategory: Autonomous Operational Governance
Type: Parent Standard
Governed Space: Autonomous Evolution
Version: 1.0
Status: Canonical · Open Standard
Origin Date: 8 July 2026
Compatibility: OOF Methodology OS · Governance Architecture (GOA™) · Operational Reality Architecture (ORA™) · Accountability Governance Architecture (AGA™) · AI Governance Architecture (AIG®) · Cognitive Governance Intelligence Architecture (CLIA®) · Memory Governance Intelligence Architecture (MGIA™)
AI-Readable: Yes
Authority: OOF®
Protection: MIP™ — Methodological Intellectual Property
Canonical Language: English (UCL)

Governed Space

Autonomous Evolution

Core Question

How may autonomous systems evolve while remaining governable?

Canonical Definition

Autonomous Evolution Standard (AEVS) defines the structural conditions under which autonomous operational systems may evolve, adapt, improve, or modify their capabilities while preserving governance integrity, operational safety, accountability, compatibility, and trustworthy autonomous operation throughout the complete evolution lifecycle.

AEVS governs autonomous evolution.

The standard establishes the governance conditions required to ensure that autonomous evolution remains authorized, controlled, validated, traceable, and continuously governable.

Autonomous systems must evolve.

Governance ensures that evolution remains trustworthy.

AEVS governs that evolution.

Governance Flow Position

Autonomous Identity → Autonomous Authority → Autonomous Execution →
Autonomous Constraints → Autonomous Risk & Trust → Autonomous
Coordination → Autonomous Escalation → Autonomous Operational
Accountability → Autonomous Compatibility → **Autonomous Evolution**

Standard Abstract

Autonomous systems cannot remain static.

Operational environments change.

Objectives change.

Threats change.

Technologies improve.

Without governed evolution:

- autonomous capabilities may become unsafe,
- operational behavior may become unpredictable,
- governance assurance declines,
- compatibility deteriorates,
- trust becomes increasingly difficult to preserve.

AEVS establishes the governance conditions required for continuous,
responsible, and governable autonomous evolution.

Operational Role

Autonomous Evolution Standard provides the governance layer
responsible for governing how autonomous operational systems evolve
without compromising operational integrity, governance legitimacy,
or accountability.

Its role is to ensure that every significant operational evolution
remains authorized, validated, traceable, and governance- approved.

Operational Space

AEVS governs:

- evolution authorization,
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- self-modification governance,
- evolution traceability,
- evolution validation,
- evolution auditability,
- governed adaptation,
- operational evolution,
- trustworthy autonomous improvement.

This governed space ensures that autonomous evolution remains compatible with governance throughout the complete operational lifecycle.

Why This Standard Exists

Long-term autonomous systems must adapt to changing operational realities.

Governance must therefore govern not only operation itself, but also operational evolution.

AEVS exists because autonomous evolution represents the **tenth governable space** of **Autonomous Systems Governance Architecture (ASGA™)**.

Scope

This standard may be applied across:

- autonomous AI agents,
 - robotics,
 - autonomous vehicles,
 - industrial automation,
 - healthcare robotics,
 - enterprise AI platforms,
 - critical infrastructure,
 - defense autonomy,
 - Human-AI operational environments,
 - distributed autonomous ecosystems.
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Relationship to Other OOF® Architectures

AEVS governs autonomous evolution.

It interoperates with:

- **GOA™** for governance principles,
 - **ORA™** for operational reality,
 - **AGA™** for accountability,
 - **AIG®** for AI behavioral governance,
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- **CLIA®** for cognition,
- **MGIA™** for memory governance.

Together these architectures ensure that autonomous systems evolve responsibly while preserving governance integrity, operational trust, accountability, and long-term operational sustainability.

Foundational Principle

Trustworthy autonomy requires governable evolution. Autonomous systems may evolve, but every significant evolution must remain authorized, validated, traceable, and continuously governable.

Canonical Closing Statement

Autonomous Evolution establishes the governance conditions required for responsible autonomous adaptation. By governing evolution itself, AEVS ensures that autonomous systems remain trustworthy, accountable, compatible, and continuously governable as they evolve throughout the complete autonomous operational lifecycle.

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Module Architecture

→ Evolution Authorization Module (EAM).

→ Self-Modification Governance Module (SMGM).

→ Evolution Traceability Module (ETM).

→ Evolution Validation Module (EVM).

→ Evolution Auditability Module (EAM).

Related Documents

→ About Autonomous Evolution Standard (AEVS).

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Canonical Source

<https://originopenfoundation.org/>